

The F-Distribution & One-Way ANOVA

1. Executive summary

- **History:** Developed by Sir Ronald A. Fisher in the 1920s.
- **Core function:** Measures the ratio of two independent sample variances.
- **Primary purpose:** Evaluates variability across multiple groups simultaneously.
- **Key metric:** The **Variance Ratio** ($F = \text{Between-Group Variance} / \text{Within-Group Variance}$).
- **Main advantage:** Controls family-wise Type I error rates across k groups.

2. Mathematical definition & key properties

- **Ratio of Chi-squares:** Ratio of two independent chi-square variables divided by their degrees of freedom.
- **Parameter pair:** Governed by numerator $df_1 = n_1 - 1$ and denominator $df_2 = n_2 - 1$.
- **Non-negative support:** Always $F \geq 0$ since variances are non-negative.
- **Asymmetric shape:** Positively skewed with a long right tail.
- **Reciprocal property:** Lower percentiles satisfy $F_{\alpha, df_1, df_2} = 1/F_{1-\alpha, df_2, df_1}$.
- **Special equivalence:** For $k = 2$ groups, the F -statistic equals Student's t^2 $F = t^2$

3. Interpretation

- **Null scenario** ($F \approx 1$):
 - Between-group variation matches background noise.
 - Group mean differences are likely due to random chance.
 - **Alternative scenario** ($F \gg 1$):
 - Between-group variation significantly exceeds random variation.
 - Signals genuine treatment or group differences.
 - **Underlying assumptions:**
 - Normally distributed populations.
 - Homogeneity of variance (equal group variances).
 - Independent sampling of observations.
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4. Scope of applications

- **Variance ratio test:** Compares two population variances ($H_0: \sigma_1^2 = \sigma_2^2$)
- **One-Way ANOVA:** Tests equality of k group means ($H_0: \mu_1 = \mu_2 = \dots = \mu_k$)
- **Advanced designs:** Applies to Randomized Complete Block, Factorial, and Repeated Measures ANOVA.
- **Multivariable regression:** Tests overall model significance ($F = MSR/MSE$) and extra sums of squares.

5. One-Way ANOVA hypothesis testing example

Study design & data

- **Context:** Testing 3 weight-loss diets ($k = 3$) across $N = 9$ subjects ($n_j = 3$ per diet).
- **Diet 1** ($Group_1$)- 6 ,5 ,4 pounds lost
- **Diet 2** ($Group_2$)- 9,8,7 pounds lost
- **Diet 3** ($Group_3$)- 12, 11,10 pounds lost

5. One-Way ANOVA hypothesis testing example

Step-by-Step calculation summary

- **Group means:** $\bar{x}_1 = 5, \bar{x}_2 = 8, \bar{x}_3 = 11$.
- **Grand mean:** $\bar{x}_{..} = (5 + 8 + 11)/3 = 8$.
- **Sum of Squares Among (SSA):** $3(5 - 8)^2 + 3(8 - 8)^2 + 3(11 - 8)^2 = 54$.
- **Among Degrees of freedom (df_A):** $k - 1 = 3 - 1 = 2$.
- **Mean Square Among (MSA):** $SSA/df_A = 54/2 = 27$.
- **Sum of Squares Within (SSW):** $(1 + 0 + 1) + (1 + 0 + 1) + (1 + 0 + 1) = 6$.
- **Within Degrees of freedom (df_W):** $N - k = 9 - 3 = 6$.
- **Mean Square Within (MSW):** $SSW/df_W = 6/6 = 1$.
- **Calculated F-Statistic:** $F = MSA/MSW = 27/1 = 27$.

5. One-Way ANOVA hypothesis testing example

ANOVA Summary table

Source of Variation	Sum of Squares (SS)	Degrees of Freedom (df)	Mean Square (MS)	F-Value
Among Groups	54	2	27	27.0
Within Groups	6	6	1	—
Total	60	8	—	—

5. One-Way ANOVA hypothesis testing example

- **Decision:** Computed $F = 27$ far exceeds the critical value $F_{0.05,2,6} = 5.14$) $P < 0.001$,(confirming statistically significant differences among the three diets.

6. Post-Hoc pairwise comparisons (Tukey HSD & Bonferroni)

- **Why Post-Hoc Testing Is Essential:**
 - An overall F -test indicates *that* group means differ, but not *which* specific pairs differ.
 - Running multiple unadjusted t -tests inflates the overall Type I error rate.
 - Post-hoc tests pinpoint specific pairwise differences while maintaining error control.
- **Tukey's Honestly Significant Difference (HSD):**
 - **Method:** Uses the Studentized Range distribution to calculate a single critical threshold for all pairwise differences.
 - **Best Used For:** All possible pairwise comparisons among group means.
 - **Key Feature:** Maintains exact family-wise error rate α without being overly conservative.
 - **In R:** Evaluated via `TukeyHSD(anova_model)`.
- **Bonferroni Adjustment:**
 - **Method:** Adjusts significance threshold to α/m (or multiplies p -values by m , where $m = k(k-1)/2$ comparisons).
 - **Best Used For:** Small set of pre-planned or custom pairwise comparisons.
 - **Key Feature:** Highly robust and versatile, but can become overly conservative as k grows.
 - **In R:** Evaluated via `pairwise.t.test(..., p.adjust.method = "bonferroni")`.

7. Complete R codes

```
# --- 1. Load Study Data ---  
diet_data <- data.frame(  
  Diet = factor(rep(c("Diet 1", "Diet 2", "Diet 3"), each = 3)),  
  WeightLoss = c(4, 5, 6, 7, 8, 9, 10, 11, 12)  
)  
  
# --- 2. Fit One-Way ANOVA Model ---  
anova_model <- aov(WeightLoss ~ Diet, data = diet_data)  
  
# --- 3. Display ANOVA Summary Table ---  
summary(anova_model)
```

```
##           Df Sum Sq Mean Sq F value Pr(>F)
## Diet       2     54      27      27 0.001 ***
## Residuals  6       6       1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

ANOVA table output

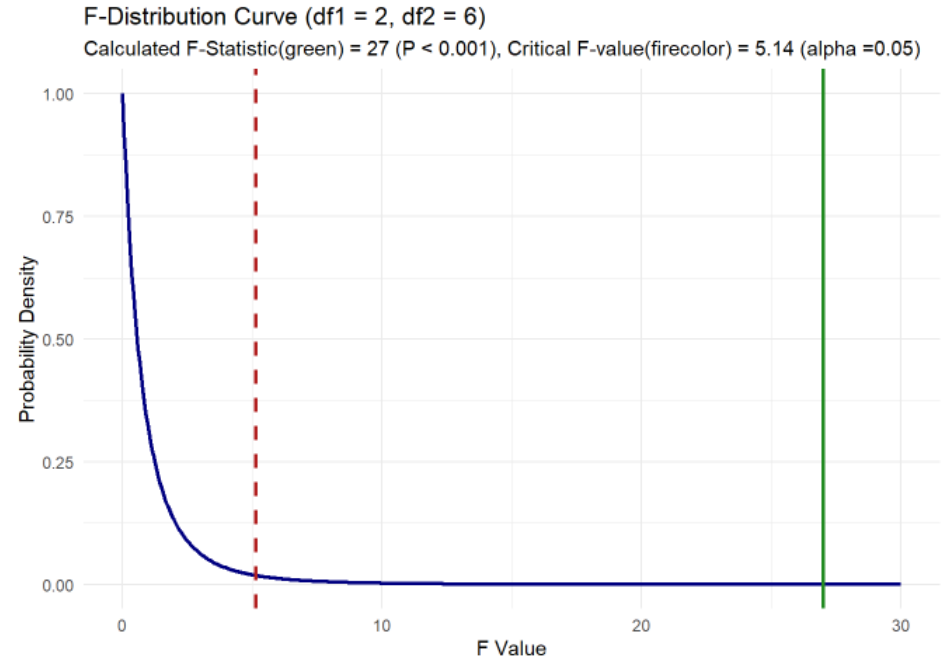
7. Complete R codes

```
# --- 4. Plot F-Distribution Curve & Test Statistic ---
f_stat <- summary(anova_model)[[1]][["F value"]][1]
df1 <- summary(anova_model)[[1]][["Df"]][1]
df2 <- summary(anova_model)[[1]][["Df"]][2]

x_vals <- seq(0, 30, length.out = 500)
y_vals <- df(x_vals, df1, df2)
plot_df <- data.frame(x = x_vals, y = y_vals)
criticalF <- qf(p = 1-0.05, df1 = 2, df2 = 6)

ggplot(plot_df, aes(x = x, y = y)) +
  geom_line(color = "navy", size = 1) +
  geom_vline(xintercept = f_stat, color = "forestgreen", linetype = "solid", size = 1) +
  geom_vline(xintercept = criticalF, color = "firebrick", linetype = "dashed", size = 1) +
  labs(
    title = "F-Distribution Curve (df1 = 2, df2 = 6)",
    subtitle = paste("Calculated F-Statistic =", f_stat, "(P < 0.001)"),
    x = "F Value",
    y = "Probability Density"
  ) +
  theme_minimal()
```

F-distribution curve



7. Complete R codes

```
# --- 5. Post-Hoc Comparisons: Tukey HSD ---
cat("\n--- Tukey HSD Pairwise Comparisons ---\n")
tukey_results <- TukeyHSD(anova_model)
print(tukey_results)

# --- 6. Post-Hoc Comparisons: Bonferroni Adjustment ---
cat("\n--- Bonferroni Adjusted Pairwise t-Tests ---\n")
bonf_results <- pairwise.t.test(
  x = diet_data$WeightLoss,
  g = diet_data$Diet,
  p.adjust.method = "bonferroni"
)
print(bonf_results)
```

Post-Hoc pairwise comparison

```
##
## --- Tukey HSD Pairwise Comparisons ---
##   Tukey multiple comparisons of means
##     95% family-wise confidence level
##
## Fit: aov(formula = WeightLoss ~ Diet, data = diet_data)
##
## $Diet
##           diff          lwr          upr          p adj
## Diet 2-Diet 1      3 0.4947644 5.505236 0.0242291
## Diet 3-Diet 1      6 3.4947644 8.505236 0.0007942
## Diet 3-Diet 2      3 0.4947644 5.505236 0.0242291
##
## --- Bonferroni Adjusted Pairwise t-Tests ---
##
##   Pairwise comparisons using t tests with pooled SD
##
## data:  diet_data$WeightLoss and diet_data$Diet
##
##           Diet 1  Diet 2
## Diet 2 0.03121 -
## Diet 3 0.00097 0.03121
##
## P value adjustment method: bonferroni
```